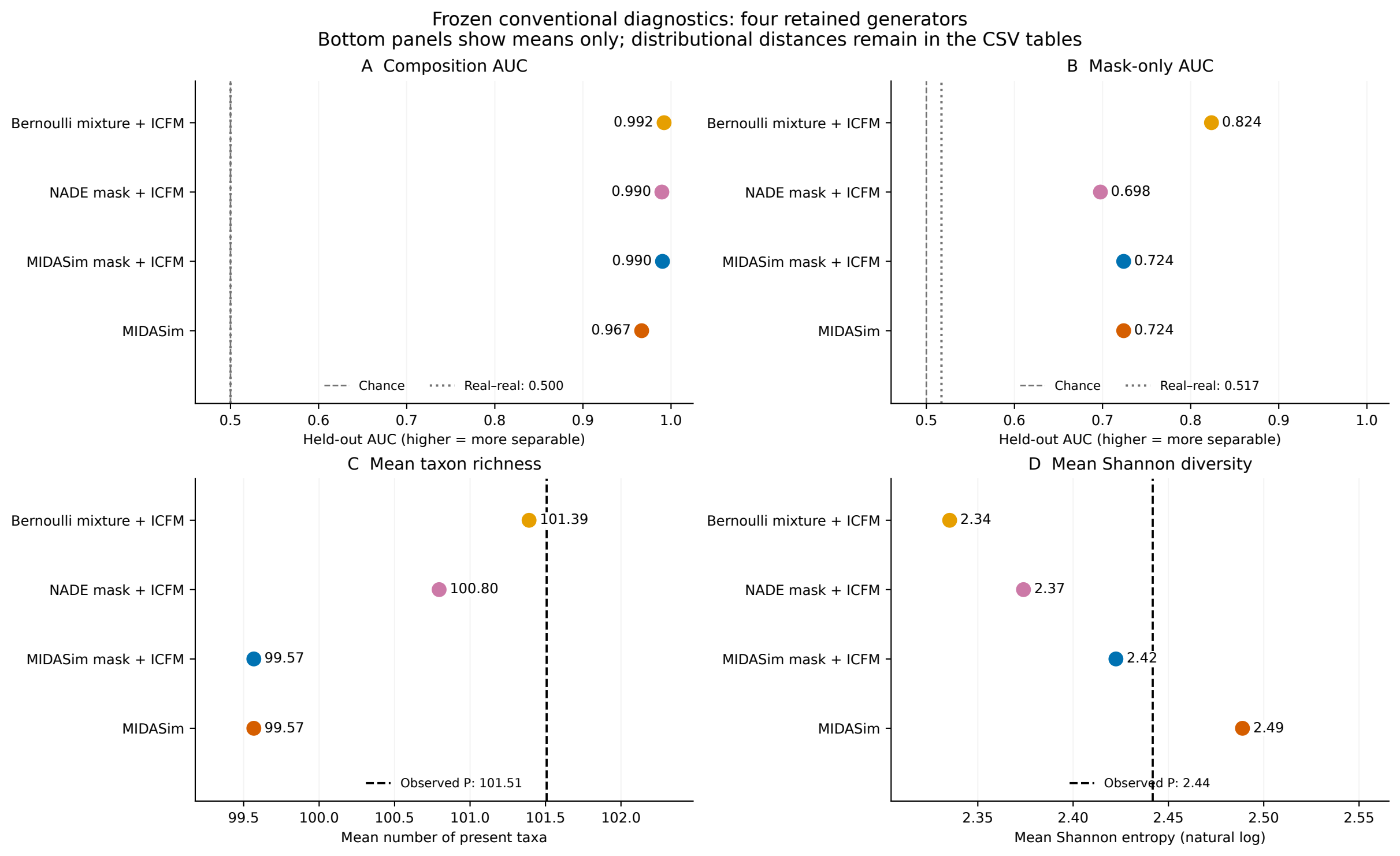
Frozen conventional diagnostics: four retained generators
Bottom panels show means only; distributional distances remain in the CSV tables
A Composition AUC
Bernoulli mixture + ICFM
NADE mask + ICFM
MIDASim mask + ICFM
MIDASim
0.992
0.990
0.990
0.967
Chance
Real–real: 0.500
0.5
0.6
0.7
0.8
0.9
1.0
Held-out AUC (higher = more separable)
B Mask-only AUC
Bernoulli mixture + ICFM
NADE mask + ICFM
MIDASim mask + ICFM
MIDASim
0.824
0.698
0.724
0.724
Chance
Real–real: 0.517
0.5
0.6
0.7
0.8
0.9
1.0
Held-out AUC (higher = more separable)
C Mean taxon richness
Bernoulli mixture + ICFM
NADE mask + ICFM
MIDASim mask + ICFM
MIDASim
101.39
100.80
99.57
99.57
Observed P: 101.51
99.5
100.0
100.5
101.0
101.5
102.0
Mean number of present taxa
D Mean Shannon diversity
Bernoulli mixture + ICFM
NADE mask + ICFM
MIDASim mask + ICFM
MIDASim
2.34
2.37
2.42
2.49
Observed P: 2.44
2.35
2.40
2.45
2.50
2.55
Mean Shannon entropy (natural log)